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Method to build a service function chain in an overlay network

Abstract

Implementations are provided for building a service function chain in a network that can remove the requirement to encapsulate packet level metadata or other packet levels in a service function chain. The approach allows for implementing service function chains while separating chain metadata from the service function so that service functions are not required to support handling service function chaining metadata. The approach can map a tunnel identifier of an encapsulated data packet to an attachment circuit of a service function and can rewrite the destination media access control (MAC) address of the data packet to the MAC address of the service function.

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Background/Summary

BACKGROUND

(1) In a network, traffic can be sent to one or more service functions for policy enforcement. In current deployment models, a service function is tightly coupled to the underlying network topology and is often implemented as an embedded service within a piece of network equipment, resulting in relatively rigid and static deployments. In modern networks, service functions can be decoupled from the underlying network topology and deployed as an ordered chain of virtualized functions, known as a service function chain, in public or private clouds. This approach requires the service infrastructure and the service functions to insert and inspect additional packet level encapsulations to convey service function chain information or metadata. This can represent a significant challenge as the service functions may not have the capability to handle additional service function chain encapsulations.

BRIEF SUMMARY

(2) Generally disclosed herein is an approach for building a service function chain in a network that can remove the requirement to encapsulate packet level metadata or other packet levels in a service function chain. The approach allows for implementing service function chains while separating chain metadata from the service function so that service functions are not required to support handling service function chaining metadata. The approach can map a tunnel identifier of an encapsulated data packet to an attachment circuit of a service function and can rewrite the destination media access control (MAC) address of the data packet to the MAC address of the service function.

(3) An aspect of the disclosure provides for a method for routing a data packet through a service function chain. The method includes encapsulating, with one or more processors, the packet with a header including a source IP address, a destination IP address of a destination service function forwarder for a service function of the service function chain, and a tunnel identifier uniquely identifying a tunnel in the service function chain. The method further includes rewriting, with the one or more processors, a destination media access control (MAC) address of the packet to a MAC address of an attachment circuit connected to the destination service function forwarder. The method also includes mapping, with the one or more processors, the tunnel identifier to the attachment circuit. The method further includes sending, with the one or more processors, the encapsulated packet to the service function forwarder through the tunnel identified by the tunnel identifier.

(4) In an example, the source IP address is a source IP address of one of a service edge node or a source service function forwarder. In another example, the method further includes setting, with the one or more processors, a source MAC address of the packet. In yet another example, the source MAC address is a source MAC address of one of a service edge node or a source service function forwarder. In yet another example, the method further includes performing, with the one or more processors, a route lookup using the destination IP address of the packet to provide the tunnel identifier and the MAC address of the attachment circuit connected to the destination service

function forwarder.

(5) In yet another example, the method further includes decapsulating, with the one or more processors, the encapsulated packet at the destination service function forwarder. In yet another example, the method further includes encapsulating, with the one or more processors, the packet with a header comprising an additional source IP address, a destination IP address of a service edge node, and a tunnel identifier uniquely identifying an additional tunnel in the service function chain. In yet another example, the method further includes rewriting, with the one or more processors, the destination MAC address of the packet to a MAC address of the service edge node. In yet another example, the method further includes mapping, with the one or more processors, the tunnel identifier uniquely identifying the additional tunnel to the service edge node. In yet another example, the method further includes sending, with the one or more processors, the encapsulated packet to the service edge node through the additional tunnel.

(6) Another aspect of the disclosure provides for a system including one or more processors; and one or more storage devices coupled to the one or more processors and storing instructions that, when executed by the one or more processors, causes the one or more processors to perform operations for routing a data packet through a service function chain. The operations include encapsulating the packet with a header comprising a source IP address, a destination IP address of a destination service function forwarder for a service function of the service function chain, and a tunnel identifier uniquely identifying a tunnel in the service function chain. The operations further include rewriting a destination media access control (MAC) address of the packet to a MAC address of an attachment circuit connected to the destination service function forwarder. The operations also include mapping the tunnel identifier to the attachment circuit. The operations further include sending the encapsulated packet to the service function forwarder through the tunnel identified by the tunnel identifier.

(7) In an example, the source IP address is a source IP address of one of a service edge node or a source service function forwarder. In another example, the operations further include setting a source MAC address of the packet, where the source MAC address is a source MAC address of one of a service edge node or a source service function forwarder. In yet another example, the operations further include performing a route lookup using the destination IP address of the packet to provide the tunnel identifier and the MAC address of the attachment circuit connected to the destination service function forwarder.

(8) In yet another example, the operations further include decapsulating the encapsulated packet at the destination service function forwarder. In yet another example, the operations further include encapsulating the packet with a header comprising an additional source IP address, a destination IP address of a service edge node, and a tunnel identifier uniquely identifying an additional tunnel in the service function chain; rewriting the destination MAC address of the packet to a MAC address of the service edge node; mapping the tunnel identifier uniquely identifying the additional tunnel to the service edge node; and sending the encapsulated packet to the service edge node through the additional tunnel.

(9) Yet another aspect of the disclosure provides for a non-transitory computer readable medium for storing instructions that, when executed by one or more processors, causes the one or more processors to perform operations for routing a data packet through a service function chain. The operations include encapsulating the packet with a header comprising a source IP address, a destination IP address of a destination service function forwarder for a service function of the service function chain, and a tunnel identifier uniquely identifying a tunnel in the service function chain. The operations further include rewriting a destination media access control (MAC) address of the packet to a MAC address of an attachment circuit connected to the destination service function forwarder. The operations also include mapping the tunnel identifier to the attachment circuit. The operations further include sending the encapsulated packet to the service function forwarder through the tunnel identified by the tunnel identifier.

- (10) In an example, the operations further include setting a source MAC address of the packet, where the source MAC address is a source MAC address of one of a service edge node or a source service function forwarder. In another example, the operations further include performing a route lookup using the destination IP address of the packet to provide the tunnel identifier and the MAC address of the attachment circuit connected to the destination service function forwarder.
- (11) In yet another example, the operations further include encapsulating the packet with a header comprising an additional source IP address, a destination IP address of a service edge node, and a tunnel identifier uniquely identifying an additional tunnel in the service function chain; rewriting the destination MAC address of the packet to an MAC address of the service edge node; mapping the tunnel identifier uniquely identifying the additional tunnel to the service edge node; and sending the encapsulated packet to the service edge node through the additional tunnel.
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Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 depicts a block diagram of an example network environment with a network device according to aspects of the disclosure.
- (2) FIG. 2 depicts a block diagram of an example service function chain domain for implementing the tunnel identifier mapping and destination MAC address rewriting according to aspects of the disclosure.
- (3) FIG. 3 depicts a block diagram of an example service function chain domain with two service functions for implementing the tunnel identifier mapping and destination MAC address rewriting according to aspects of the disclosure.
- (4) FIG. 4 depicts a flow diagram of an example process for routing data packets through a SFC in a SFC domain according to aspects of the disclosure.
- (5) FIG. 5 depicts a block diagram of an example computing system according to aspects of the disclosure.

DETAILED DESCRIPTION

- (6) Generally disclosed herein is an approach for building a service function chain in a network, where a tunnel identifier of an encapsulated data packet is mapped to an attachment circuit of the service function and a destination media access control (MAC) address of the data packet is rewritten to the MAC address of the service function. The approach can enable a service function chain implementation while abstracting chain metadata to allow each service function to remain oblivious to underlying chain implementations.
- (7) FIG. 1 depicts an example network environment **100** with a network device **110**. The network environment **100** includes a network **120** of interconnected network nodes **125**. The network nodes **125** participate in the network **120** as data sources, data destinations or data sinks, and/or intermediary nodes such as switches, routers, and/or gateways propagating data from sources towards destinations through the network **120**. The network nodes **125** can be service functions, service function forwarders, and/or service edge nodes, to be described further below with respect to FIGS. 2-3. The network **120** includes the network device **110** with links **127** to various other participating network nodes **125**.
- (8) The network **120** can facilitate interactions between participant devices. Example networks include the Internet, a local network, a network fabric, or any other local area or wide area network. The network **120** can be composed of multiple connected sub-networks or autonomous networks. The network **120** can be a local-area network (LAN), such as a company intranet, a metropolitan area network (MAN), a wide area network (WAN), an inter-network such as the Internet, or a peer-to-peer network. Any type and/or form of data network and/or communication network can be used for the network **120**. It can be public, private, or a combination of public and private networks. In

general, the network **120** can be used to convey information between computing devices; for example, network nodes **125**, and the network device **110**.

(9) The network device **110** can be a host device or server executing one or more virtual machines **130** on one or more CPUs of the network device **110**. The network device **110** can include a network interface driver **140**, a memory **150**, a network interface **160**, a real operating system (OS) **170**, a hypervisor **180**, and the virtual machines **130**. The network device **110** can communicate with other network devices over the network **120**. The network device **110** and other network devices can create or participate in one or more virtual networks to facilitate communication between virtual machines **130**. The one or more virtual networks can overlay the physical network that makes up the network **120**. Routing data packets within one of the virtual networks overlaying the physical network can be accomplished using a system of data packet encapsulation, to be described further below with respect to FIGS. 2-3. The network device **110** and the network nodes **125** can route the data packets according to virtual Internet protocol addresses (VIPs) and MAC addresses. In some examples, the network device **110** can host one or more virtual machines **130** that transmit data packets or receive data packets. In other examples, the network device **110** can be an intermediate node of a path between two virtual machines executing on two separate network devices. Accordingly, the network device **110** can act as a source, destination, or intermediate switch in one or more virtual networks.

(10) The hypervisor **180** can manage operation of the virtual machines **130**, including spinning up and terminating the virtual machines **130**, managing allocation of memory **150** to the virtual machines **130**, and live migrating the virtual machines **130** to other network devices.

(11) The memory **150** can store data and/or computer executable instructions related to the operation and use of the network interface driver **140**. The memory **150** can include a random access memory (RAM), a dynamic random access memory (DRAM), a static random access memory (SRAM), a synchronous dynamic random access memory (SDRAM), a ferroelectric random access memory (FRAM), a read only memory (ROM), a programmable read only memory (PROM), an erasable programmable read only memory (EPROM), an electrically erasable programmable read only memory (EEPROM), and/or a flash memory, as examples.

(12) The memory **150** can store computer executable instructions of a transport protocol module **175**, such as a transmission control protocol (TCP) module or the TCP layer of a network stack, to be executed on a processor of the network device **110**. Functionality occurring within a TCP layer of a network stack can be additionally or alternatively executed in another network protocol module within the transport layer, the network layer, or a combined transport/network layer of a network stack. For example, the functionality can be implemented in a user datagram protocol (UDP) module, reliable datagram protocol (RDP) module, reliable user datagram protocol (RUDP) module, or a datagram congestion control protocol (DCCP) module. A network layer, a transport layer, or a combined transport/network layer can generally be referred to as a packet layer of a network stack.

(13) The network interface driver **140** can include a network interface driver software module running on the real OS **170**. As an example, the network interface driver can be a collection of computer executable instructions stored in the memory **150** that when executed by a processor help facilitate network communications. As another example, the network interface driver **140** can be implemented as logic in a hardware processor or other integrated circuit, or as a combination of hardware and software logic. The network interface driver **140** can communicate with one of the software virtual machines **130** directly or via a guest OS of the virtual machine **130** and/or the hypervisor **180**.

(14) The network interface driver **140** can be included within a layer of a network stack of the real OS **170** of the network device **110**. In an example, the network interface driver **140** can be included within a data link layer of a network stack and can communicate with a software module or application that is included in an application layer of the network stack. In another example, the

network interface driver **140** can be included within a link layer of a network stack and can communicate with a TCP/IP module that is included in an internet/transport layer of the network stack. In other examples, the functionality occurring within the layer of the network stack can additionally or alternatively be configured to receive packets from another network or transport layer protocol module, such as a user datagram protocol (UDP) module, reliable datagram protocol (RDP) module, reliable user datagram protocol (RUDP) module, or a datagram congestion control protocol (DCCP) module. While not shown, the network interface driver **140** can be included as a portion of the network interface **160**.

(15) The network interface **160** can include a packet forwarder **162**, an address matcher **164**, a connection tracker **166**, and a memory **168**. The memory **168** can store data and/or computer executable instructions related to the operation and use of the network interface card **160**. The memory **168** can store route lookup tables that include destination addresses corresponding to VIP and MAC addresses. The memory **168** can include a random access memory (RAM), a dynamic random access memory (DRAM), a static random access memory (SRAM), a synchronous dynamic random access memory (SDRAM), a ferroelectric random access memory (FRAM), a read only memory (ROM), a programmable read only memory (PROM), an erasable programmable read only memory (EPROM), an electrically erasable programmable read only memory (EEPROM), and/or a flash memory, as examples.

(16) The packet forwarder **162** can be implemented in an application-specific integrated circuit (ASIC) or as software executing on a microprocessor. The packet forwarder **162** can receive packets for transmission from the virtual machines **130** and route them to one or more destination virtual machines executing on remote network devices. The packet forwarder **162** can encapsulate a data packet with a header that includes a source IP address, a destination IP address, and a tunnel identifier, rewrite a destination MAC address of the data packet, map the tunnel identifier, and transmit the encapsulated data packet. The packet forwarder **162** can also perform functions that allow the network interface **160** to act as a destination for data packets received over the links **127**. For example, the packet forwarder **162** can receive a data packet over one of the lines **127**, decapsulate the data packet, and forward the data packet to one or more destination virtual machines **130** executing on the network device **110**.

(17) The address matcher **164** can be implemented in an application-specific integrated circuit (ASIC) or as software executing on a microprocessor. As an example, the address matcher **164** can receive a data packet from a local virtual machine, i.e., a virtual machine **130** executing on the local network device **110**, where the destination for the data packet is a virtual machine executing on a remote network device. The address matcher **164** can determine the destination virtual machine of the data packet based on the longest prefix match of a destination address of the data packet, where the destination address corresponds to the destination virtual machine. The address matcher **164** can select a path to the destination network device from a plurality of equal-cost multipath (ECMP) paths. The address matcher **164** can then encapsulate the data packet with an outer header having a second destination address that corresponds to the destination network device. The address matcher **164** can then transmit the encapsulated packet.

(18) The connection tracker **166** can be implemented in an application-specific integrated circuit (ASIC) or as software executing on a microprocessor. As an example, the connection tracker **166** can perform firewall type functions whereby it monitors packets for transmission from the virtual machines **130** executing on the network device **110** destined for target virtual machines, hosts, or clients external to the network device **110**. The connection tracker **166** can determine whether either or both of a source port number or a destination port number fall within a range of port numbers permitted by the policies imposed on the network device **110** and/or the virtual machines **130**. In addition, the connection tracker **166** can record metadata regarding the data packet in a connection flow table or other log maintained in the network interface memory **168**. In this manner, the connection tracker **166** can check port numbers and record log entries in hardware on the

network interface **140**; that is, between the processor of the network interface **160** and the memory **168**. If the connection tracker **166** determines that one of the port numbers is not within a permitted range, it can drop or block the data packet and can also send a notification to the virtual machine **130** and/or the hypervisor **180** notifying it of a possible breach of a network policy. If the port number is determined to be within range, the connection tracker **166** can insert an entry corresponding to the data packet in the connection flow table in the memory **168** and transmit the data packet.

(19) FIG. 2 depicts an example service function chain (SFC) domain **200** for implementing the tunnel identifier mapping and destination MAC address rewriting. The SFC domain **200** can be a network, such as an overlay network. An overlay network can be a virtualized network supported by an underlying physical network. Example overlay networks can include distributed systems such as peer-to-peer networks and client-server applications. The network can be the network **120**, as described with respect to FIG. 1.

(20) The SFC domain **200** can include a plurality of service functions (SFs) **210** that can form a service function chain (SFC). The SFC can include any number of SFs **210**. A SFC defines an ordered set of SFs and steers traffic of data packets therethrough. One or more SFs can be applied to the data packet or frame in a specific order.

(21) A SF **210** can be responsible for a specific treatment of a received data packet. The SF **210** can be a virtual element or be embedded in a physical network element. Example SFs **210** can include wide area network and application accelerating, encapsulation/decapsulation, stateful or stateless packet filtering, intrusion detection, network address translation (NAT), lawful intercept, and transmission control protocol (TCP) optimization.

(22) Each SF **210** can be connected to two service function forwarders (SFFs) **220** via respective attachment circuits **230**. A SFF **220** can be responsible for forwarding packets to one or more connected SFs **210** based on information carried in the packet or explicit SFC encapsulation. A SFF **220** can also be responsible for handling packets returning from the SF **210** and delivering packets to other SFFs **220**.

(23) The attachment circuit **230** can be an interface construct that attaches a SF **210** to an SFF **220**. There can be at least one attachment circuit **230** for a SF **210**. For example, a terminating SF can have one attachment circuit while a transit SF can have two attachment circuits, which can be referred to as left and right or first and second. Forward traffic can enter the SF on its left or first interface and leave on the right or second interface with the SF applied. Return traffic can enter the SF on its right or second interface and leave on the left or first interface with the SF applied.

(24) The SFFs **220** of different SFs **210** can be connected to each other via tunnels **240**. Encapsulated data packets are transmitted through the tunnels **240** to their destinations.

(25) The SFC can further include at least one service edge node **250** for entering and/or exiting the SFC domain **200**. The service edge node **250** can be the same node for entering and exiting if the node supports multiple lookup tables, e.g., a lookup table for entry and a lookup table for exit. Otherwise, the SFC can include a service edge node **250** for entering the domain and a service edge node **250** for exiting the domain. The service edge nodes **250** can each be connected to a SFF **220** via a tunnel **240**. A SFF **220** can further be responsible for handling data packets received from a service edge node **250** or delivering data packets to a service edge node **250**.

(26) The service edge nodes **250** can be connected to a network. For example, the service edge node **250** for entering the domain can be connected to an internal or local network **260** while the service edge node for exiting the domain can be connected to the Internet **270**.

(27) A data packet from the network **260** can be sent to the SFC domain **200** through a service edge node **250**. The data packet can be forward traffic or return traffic. The domain **200** can be bi-directional or unidirectional.

(28) The service edge node **250** can encapsulate the packet, including setting a tunnel identifier for determining through which tunnel **240** to send the encapsulated packet, and send the encapsulated

packet to a SFF **220** of a SF **210** through the identified tunnel **240**. A tunnel identifier can uniquely identify each tunnel **240** in the SFC. Example tunnel identifiers include virtual network identifiers (VNIs) such as a VXLAN VNI, or a GENEVE VNI. On the service edge node **250**, a route lookup can be performed using a destination IP address of the packet. The result of the lookup can provide a tunnel identifier and a MAC address of an attachment circuit **230** attaching the SFF **220** to the SF **210**.

(29) A source MAC address of the packet can be set to the MAC address of the service edge node **250** and a destination MAC address of the packet can be rewritten to the MAC address of the attachment circuit **230**. In another example, the destination MAC address of the service edge node **250** can be rewritten to the MAC address of the SFF **220**. In this example, the SFF **220** can then further rewrite the destination MAC address of the service edge node **250** to the MAC address of the SF **210**. In yet another example, the destination MAC address of the service edge node **250** can be rewritten to the MAC address of the SF **210**. In this example, the SFF **220** can forward the packet to the SF **210** without rewriting its own MAC address.

(30) Setting the source MAC address and rewriting the destination MAC address can be performed at the ingress or egress of the tunnel **240**. The packet can be encapsulated using tunnel encapsulation with tunnel source being the IP address of the service edge node **250**, destination being the address of the SFF **220**, and the tunnel identifier set. The encapsulated packet can then be sent to the SFF **220**.

(31) The SFFs **220** can map the tunnel identifier to the attachment circuit **230**. The mapping can be implemented as a direct mapping or using a route lookup. On receiving the packet over the tunnel **240**, the SFF **220** can perform a lookup using the tunnel identifier as a key and a resultant attachment circuit **230** is used to send the packet to the SF **210** after decapsulation. A SFF **220** can receive the packet from a tunnel **240**, decapsulate the packet, and forward the packet to an attachment circuit **230** to be inspected by a SF **210**. Another SFF **220** can receive the packet from an attachment circuit **230** after being inspected by an SF **210** and tunnel the packet to a different SFF **220** if the packet should be inspected by another SF **210** or to another service edge node **250** if the packet should exit the domain **200**.

(32) The SFFs **220** can also respond to address resolution protocol (ARP)/neighbor discovery (ND) requests from a connected SF **210** so that the SF **210** can remain unaware of SFC encapsulations. For example, the SFF **220** can respond to an ARP/ND request from the SF **210** to receive the packet from it, where the SF **210** can send the packet to the SFF **220** without any knowledge about the next SF **210** to process the packet or the SFC more generally.

(33) Once the packet is inspected by each SF **210** in the SFC, a SFF **220** sends the packet through a tunnel **240** to a service edge node **250**, which forwards the packet to the network **270**. Whenever the packet is sent through a tunnel **240**, the destination MAC of the packet is rewritten based on the next hop.

(34) FIG. 3 depicts an example SFC domain **300** with two SFs for implementing the tunnel identifier mapping and destination MAC address rewriting. Here, the SFC domain **300** includes a SFC formed by a first SF **302** and a second SF **304**. The first SF **302** is connected to a first SFF **306** and a second SFF **308** via a first left attachment circuit **310** and a first right attachment circuit **312**, respectively. The second SF **304** is connected to a third SFF **314** and a fourth SFF **316** via a second left attachment circuit **318** and a second right attachment circuit **320**, respectively. The SFC further includes a first service edge node **322** and a second service edge node **324**. The first service edge node **322** is connected to the first SFF **306** via a first tunnel **326**, the second SFF **308** is connected to the third SFF **314** via a second tunnel **328**, and the fourth SFF **316** is connected to the second service edge node **324** via a third tunnel **330**. The first service edge node **322** can be connected to a first network **332**, such as an internal network, and the second service edge node **324** can be connected to a second network **334**, such as the Internet.

(35) FIG. 4 depicts a flow diagram of an example process **400** for routing data packets through a

SFC in a SFC domain. The example process **400** can be performed on a system of one or more processors in one or more locations, such as the example network device **110** of FIG. **1**. The example process **400** will be described with respect to the example SFC domain **300** of FIG. **3**, but it should be noted the example process **400** can be applied to the more general SFC domain **200** of FIG. **2**. While the process **400** will describe forward traffic through the SFC domain **300**, it should also be noted the same process can be utilized for return traffic, just in reverse.

(36) As shown in block **405**, the first service edge node **322** can receive a data packet from the first network **332** and can forward the data packet to the SFC domain **300**. This can be the result of traffic classification; for example, only a subset of traffic can be sent to the first service edge node **322**.

(37) As shown in block **410**, the first service edge node **322** can perform a route lookup using the destination IP address of the data packet. The result of this lookup can provide a tunnel identifier for the first tunnel **326** and the address of the first SFF **306** that is attached to the first left attachment circuit **310** of the first SF **302**. The first service edge node **322** can set the source MAC address of the data packet to the MAC address of the first service edge node **322** and can rewrite the destination MAC address of the data packet to the MAC address of the first left attachment circuit **310**. The first service edge node **322** can encapsulate the data packet using tunnel encapsulation with the tunnel source being the IP address of the first service edge node **322**, the tunnel destination being the IP address of the first SFF **306**, and the identifier being the tunnel identifier of the first tunnel **326**.

(38) As shown in block **415**, the first service edge node **322** can send the encapsulated data packet to the first SFF **306**.

(39) As shown in block **420**, the first SFF **306** can perform a route lookup using the tunnel identifier of the first tunnel **326** as the key and the resultant attachment circuit can be used to send the data packet to the first SF **302**. The first SFF **306** can decapsulate the data packet and send the data packet to the first SF **302** via the first left attachment circuit **310**.

(40) As shown in block **425**, the first SF **302** can receive the data packet on the first left attachment circuit **310**, apply the service function, and send the data packet out on the first right attachment circuit **312**.

(41) As shown in block **430**, the second SFF **308** can receive the data packet from the first right attachment circuit **312**. The second SFF **308** can perform a route lookup using the first right attachment circuit identifier as the key. The result of this route lookup can provide a tunnel identifier and the IP address of the third SFF **314** that is attached to the left attachment circuit **318** of the second SF **304**. The second SFF **308** can set the source MAC address of the data packet to the MAC address of the second SFF **308** and can rewrite the destination MAC address of the data packet to the MAC address of the second left attachment circuit **318**. The second SFF **308** can encapsulate the data packet using the tunnel encapsulation with tunnel source being the IP address of the second SFF **308**, destination being the IP address of the third SFF **314**, and the identifier being set to the tunnel identifier of the second tunnel **328**.

(42) As shown in block **435**, the second SFF **308** can send the encapsulated data packet to the third SFF **314**.

(43) As shown in block **440**, on receiving the data packet over the second tunnel **328**, the third SFF **314** can perform a route lookup using the tunnel identifier of the second tunnel **328** as the key and the resultant attachment circuit can be used to send the data packet to the second SF **304**. The SFF **314** can decapsulate the data packet and send the data packet to the second SF **304** via the second left attachment circuit **318**.

(44) As shown in block **445**, the second SF **304** can receive the data packet on the second left attachment circuit **318**, apply the service function, and send the data packet out on the second right attachment circuit **320**.

(45) As shown in block **450**, the fourth SFF **316** can receive the data packet from the second right

attachment circuit **320**. The fourth SFF **316** can perform a route lookup using the second right attachment circuit **320** as the key. The result of this route lookup can provide a tunnel identifier and IP address of the second service edge node **330** that is attached to the second network **334**. The fourth SFF **316** can set the source MAC address of the data packet to the MAC address of the fourth SFF **316** and can rewrite the destination MAC address of the data packet to the MAC address of the second service edge node **324**. The fourth SFF **316** can encapsulate the data packet using tunnel encapsulation with the tunnel source being the IP address of the fourth SFF **316**, the tunnel destination being the IP address of the second service edge node **324**, and the identifier being the tunnel identifier of the third tunnel **330**.

(46) As shown in block **455**, the fourth SFF **316** can send the encapsulated data packet to the second service edge node **324**.

(47) As shown in block **460**, the second service edge node **324** can decapsulate the data packet, perform a route lookup using the destination IP address of the data packet, and forward the data packet to the second network **334**.

(48) FIG. 5 depicts an example computing system **500**, which can be configured as the network device **110** of FIG. 1. The computing system **500** can include one or more processors **510** in communication with memory **520**, at least one network interface controller **550** with network interface port **560** for connection to a network, input/output (I/O) components **570**, and other components **580**. The processors **510** can be in communication with the memory **520**, network interface controller **550**, I/O components **570**, and other components **580** via a bus **540**, as an example. The processors **510** can incorporate, or are connected to, cache memory **530**. In some instances, instructions are read from memory **520** into cache memory **530** and executed by the processors **510** from cache memory **530**.

(49) The processors **510** can be any logic circuitry that executes instructions fetched from the memory **520** or cache **530**. The processors **510** can be microprocessor units or special purpose processors. The computing device **500** can be based on any processor, or set of processors, capable of operating as described herein. The processors **510** can be single core or multi-core processors. The processors **510** can be multiple distinct processors. The processors **510** can be implemented as circuitry on one or more chips.

(50) The memory **520** can be any device suitable for storing computer readable data. The memory **520** can be a device with fixed storage or a device for reading removable storage media. Examples include all forms of non-volatile memory, media and memory devices, semiconductor memory devices, e.g., EPROM, EEPROM, SDRAM, and flash memory devices, magnetic disks, magneto-optical disks, and optical discs, e.g., CD ROM, DVD-ROM, or Blu-Ray® discs. The computing system **500** can have any number of memory devices **520**.

(51) The cache memory **530** can be a form of computer memory placed in close proximity to the processors **510** for fast access times. The cache memory **530** can be part of, or on the same chip as, the processors **510**. There can be multiple levels of cache **530**, e.g., level 2 and level 3 cache layers.

(52) The network interface controller **550** can manage data exchanges via the network interface **560**. The network interface controller **550** can handle the physical and data link layers of the open systems interconnection (OSI) model for network communication. Some of the tasks of the network interface controller **550** can be handled by one or more of the processors **510**. The network interface controller **550** can be incorporated into the processors **510**, e.g., as circuitry on the same chip.

(53) The computing system **500** can have multiple network interfaces **560** controlled by a single controller **550** or multiple network interface controllers **550**. Each network interface **560** can be a connection point for a physical network link, e.g., a cat-5 Ethernet link. The network interface controller **550** can support wireless network connections and an interface port **560** can be a wireless, e.g., radio, receiver/transmitter. The network interface controller **550** can implement one or more network protocols such as Ethernet. The computing system **500** can exchange data with

other computing systems via physical or wireless links through a network interface **560**. The network interface **560** can link directly to another device or to another device via an intermediary device, e.g., a network device such as a hub, a bridge, a switch, or a router, connecting the computing device **500** to a data network such as the Internet.

(54) The computing system **500** can include, or provide interfaces for, one or more input or output (I/O) components **570**. Input devices can include keyboards, microphones, touch screens, sensors, and pointing devices, such as a mouse or trackball, as examples. Output devices can include video displays, speakers, and printers, as examples.

(55) Other components **580** can include an I/O interface, external serial device ports, and any additional co-processors. For example, the computing system **500** may include an interface, e.g., a universal serial bus (USB) interface, for connecting input devices, output devices, or additional memory devices, e.g., portable flash drive or external media drive. The computing device **500** can include an additional co-processor, such as a math co-processor to assist the processors **510** with high precision or complex calculations.

(56) As such, generally disclosed herein is an approach for building a service function chain in a network that can remove the requirement to encapsulate packet level metadata or other packet levels in a service function chain. The approach allows for implementing service function chains while separating chain metadata from the service function so that service functions are not required to support handling service function chaining metadata. The approach can map a tunnel identifier of an encapsulated data packet to an attachment circuit of a service function and can rewrite the destination media access control (MAC) address of the data packet to the MAC address of the service function.

(57) Unless otherwise stated, the foregoing alternative examples are not mutually exclusive, but may be implemented in various combinations to achieve unique advantages. As these and other variations and combinations of the features discussed above can be utilized without departing from the subject matter defined by the claims, the foregoing description of the embodiments should be taken by way of illustration rather than by way of limitation of the subject matter defined by the claims. In addition, the provision of the examples described herein, as well as clauses phrased as “such as,” “including” and the like, should not be interpreted as limiting the subject matter of the claims to the specific examples; rather, the examples are intended to illustrate only one of many possible embodiments. Further, the same reference numbers in different drawings can identify the same or similar elements.

Claims

1. A method for routing a data packet through a service function chain, the method comprising: performing, with one or more processors, a route lookup using a destination internet protocol (IP) address of the data packet to determine an IP address of a service function forwarder for a service function of the service function chain, a tunnel identifier uniquely identifying a tunnel in the service function chain, and a media access control (MAC) address of an attachment circuit connected to the service function forwarder; encapsulating, with the one or more processors, the data packet with a header comprising the IP address of the service function forwarder and the tunnel identifier; rewriting, with the one or more processors, a destination MAC address of the data packet to be the MAC address of the attachment circuit; mapping, with the one or more processors, the tunnel identifier to the attachment circuit; and sending, with the one or more processors, the encapsulated data packet to the service function forwarder through the tunnel identified by the tunnel identifier.
2. The method of claim 1, wherein the header further comprises a source IP address that is of one of a service edge node or a source service function forwarder for a service function of the service function chain.

3. The method of claim 1, further comprising setting, with the one or more processors, a source MAC address of the data packet.
4. The method of claim 3, wherein the source MAC address is a source MAC address of one of a service edge node or a source service function forwarder.
5. The method of claim 1, further comprising decapsulating, with the one or more processors, the encapsulated data packet at the service function forwarder.
6. The method of claim 1, further comprising encapsulating, with the one or more processors, the data packet with a header comprising a destination IP address of a service edge node and a tunnel identifier uniquely identifying an additional tunnel in the service function chain.
7. The method of claim 6, further comprising rewriting, with the one or more processors, the destination MAC address of the data packet to be an MAC address of the service edge node.
8. The method of claim 6, further comprising mapping, with the one or more processors, the tunnel identifier uniquely identifying the additional tunnel to the service edge node.
9. The method of claim 6, further comprising sending, with the one or more processors, the encapsulated data packet to the service edge node through the additional tunnel.
10. A system comprising: one or more processors; and one or more storage devices coupled to the one or more processors and storing instructions that, when executed by the one or more processors, causes the one or more processors to perform operations for routing a data packet through a service function chain, the operations comprising: performing a route lookup using a destination internet protocol (IP) address of the data packet to determine an IP address of a service function forwarder for a service function of the service function chain, a tunnel identifier uniquely identifying a tunnel in the service function chain, and a media access control (MAC) address of an attachment circuit connected to the service function forwarder; encapsulating the data packet with a header comprising the IP address of the service function forwarder and the tunnel identifier; rewriting a destination MAC address of the data packet to be the MAC address of the attachment circuit; mapping the tunnel identifier to the attachment circuit; and sending the encapsulated data packet to the service function forwarder through the tunnel identified by the tunnel identifier.
11. The system of claim 10, wherein the header further comprises a source IP address that is of one of a service edge node or a source service function forwarder for a service function of the service function chain.
12. The system of claim 10, wherein the operations further comprise setting a source MAC address of the data packet, wherein the source MAC address is a source MAC address of one of a service edge node or a source service function forwarder.
13. The system of claim 10, wherein the operations further comprise decapsulating the encapsulated data packet at the service function forwarder.
14. The system of claim 10, wherein the operations further comprise: encapsulating the data packet with a header comprising a destination IP address of a service edge node and a tunnel identifier uniquely identifying an additional tunnel in the service function chain; rewriting the destination MAC address of the data packet to be an MAC address of the service edge node; mapping the tunnel identifier uniquely identifying the additional tunnel to the service edge node; and sending the encapsulated data packet to the service edge node through the additional tunnel.
15. A non-transitory computer readable medium for storing instructions that, when executed by one or more processors, causes the one or more processors to perform operations for routing a data packet through a service function chain, the operations comprising: performing a route lookup using a destination internet protocol (IP) address of the data packet to determine an IP address of a service function forwarder for a service function of the service function chain, a tunnel identifier uniquely identifying a tunnel in the service function chain, and a media access control (MAC) address of an attachment circuit connected to the service function forwarder; encapsulating the data packet with a header comprising the IP address of the service function forwarder and the tunnel identifier; rewriting a MAC address of the data packet to be the MAC address of the attachment

circuit; mapping the tunnel identifier to the attachment circuit; and sending the encapsulated data packet to the service function forwarder through the tunnel identified by the tunnel identifier.

16. The non-transitory computer readable medium of claim 15, wherein the operations further comprise setting a source MAC address of the data packet, wherein the source MAC address is a source MAC address of one of a service edge node or a source service function forwarder.

17. The non-transitory computer readable medium of claim 15, wherein the operations further comprise: encapsulating the data packet with a header comprising a destination IP address of a service edge node and a tunnel identifier uniquely identifying an additional tunnel in the service function chain; rewriting the MAC address of the data packet to be an MAC address of the service edge node; mapping the tunnel identifier uniquely identifying the additional tunnel to the service edge node; and sending the encapsulated data packet to the service edge node through the additional tunnel.
